

# A fixed-time obstruction to PAH-v2 semigroup comparison

TECT verification-first repository  
claim C6-SPACETIME-SIGNATURE

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## 1. Purpose and scope: the unchanged target

Put  $f_r = I_{r_0,r}f_0$  and  $S_r(t) = \exp(tL_r)$ , where  $L_r$  is the full original PAH-v2 finite row generator on  $\mathbb{C}^{\Omega_r}$ , restricted to the invariant algebra when appropriate. PAH-V2-SG-001 asks whether

$$\forall T < \infty, \quad \lim_{R \rightarrow \infty} \sup_{s > r} \sup_{R \leq t \leq T} \|D_{r,s}(t)\|_{L^2(\pi_s)} = 0, \quad D_{r,s}(t) = S_s(t)I_{r,s}f_r - I_{r,s}S_r(t)f_r. \quad (1)$$

All states, parameters, maps and time units are those of R-573. Its generator counterexample alone did not settle (1). Here we supply the missing uniform finite-time estimate; no infinite semigroup is presumed to exist.

The immutable preregistration, including  $t_* = 1/10$  before computation, is

strategy/pa-hyp/PAH-v2-SG-001-prereg-v1.json  
177e36c3ea7370f2377c9a53b385396e7b6de1a28399c3b306880c92ce672646

It pins PAH-001-v1 (functional/action and dynamics), PAH-001-v2-r2 (root partial maps and operators/pairings), R-570 finite operators, R-571 comparison admission, the Morton/CRT bundle and R-573's execution and proof. The adopted comparison-bundle hash remains

1d6fa47f5351d74f05bed3aba5b1e229a827644d4c6ef5d7dd73c117adee95b6

The result certificate lists all source and evidence hashes.

The carrier is the existing  $h = N = 0$  anchored open square: four vertices, four stored edges and one bounded face;  $O = (0, 0)$  and  $C = (1, 0)$  are labels. These are two-dimensional combinatorial addresses, not physical dimensions. Fix  $r_0 = 2$ ,  $q_0 = m_0 = R_0 = 1$ , all original rate scales equal to one, and

$$\epsilon = \beta = \nu = \lambda_s = \eta_6 = \kappa_s = \kappa_D = \kappa_g = g = 1, \quad m^2 = \lambda_4 = 0. \quad (2)$$

The original schedule is  $K_r = \prod_{i=1}^{r+1} p_i$ ,  $M_{s,r} = 2^r$ ,  $M_{\psi,r} = 4^r$ ,  $R_{\max,r} = 2^r$ ,  $Q_r = 2^r$ . All full counting labels remain distinct. The original weight is  $\pi_r(x) = e^{-F_r(x)} / \sum_{y \in \Omega_r} e^{-F_r(y)}$ . Only the local-state cutoff tends to infinity; later lattice  $h$ , volume  $N$  and inherited limit stages are unchanged and unexecuted. Time remains the original external stochastic Markov time and stationary initial-value boundary.

## 2. Exact source-to-character crosswalk

For the signed face sum  $\Phi = \sum_{e \in \partial p} \sigma_e u_e$ , set

$$f_0 = e^{2\pi i \Phi/5}, \quad a_r = (K_r/K_2)^{-1} \pmod{5}, \quad f_r = e^{2\pi i a_r \Phi/5}. \quad (3)$$

Closed-face gauge coboundaries telescope, and modular wrap is invisible since  $5 \mid K_r$ . The full cell-automorphism group fixing the two adjacent labelled anchors is trivial. Thus (3) is in the unchanged

full invariant algebra. R-571's original inverse-prime projection gives exactly these pullbacks,  $I_{r,s}f_r = f_s$ , and  $|f_r| = 1$  on every original state. The finite domain is the entire function space; R-570's symmetry commutation preserves its invariant part. No new physical projection is used.

R-573, sections 2–4, establishes for the same full generator

$$L_r f_r = \lambda_r f_r + e_r, \quad \lambda_r \in \{-10 - 2\sqrt{5}, -10 + 2\sqrt{5}\}, \quad \|e_r\|_\infty \leq 256/K_r. \quad (4)$$

For completeness, the changing energy under a signed LK move is at most  $4\pi/K_r \leq 16/K_r$ : one matter edge and its sole face change, while  $A_v = 2^{-r}\ell_v \geq 0$  and  $\sum_v A_v = 1$ . The original rate is  $e^{-\Delta F/2}$ , so for  $K_r \geq 8$  its difference from one is at most  $16/K_r$ , using  $e^u - 1 \leq 2u$  for  $0 \leq u \leq 1$ . Eight signed incidences times the character increment bound two give  $256/K_r$ . The PH/TR/AP increments of the initial character vanish individually, and the unit-rate LK leading term is  $4(2\cos(2\pi a_r/5) - 2)$ . This is a decomposition, not replacement rates. PH/TR/AP are retained in the full evolved semigroup; no invariant one-character span is asserted.

### 3. Full finite Markov contraction and the time-error estimate

At each finite  $r$  the original off-diagonal rates are nonnegative and finite, and the row sum is zero. Let  $M_r = 1 + \max_x \sum_{y \neq x} q_r(x, y)$ . The matrix  $P_r = 1 + L_r/M_r$  is stochastic. The exact matrix exponential is

$$S_r(t) = e^{-M_r t} \sum_{n=0}^{\infty} \frac{(M_r t)^n}{n!} P_r^n.$$

The coefficients sum to one. Triangle inequalities in each row therefore give  $\|S_r(t)v\|_\infty \leq \|v\|_\infty$  for complex  $v$  and all  $t \geq 0$ .  $M_r$  need not be uniform: it is an algebraic proof device, not a changed rate or time scale. This proves contraction for all original roots at once.

Differentiate  $S_r(t-u)e^{\lambda_r u} f_r$  for  $0 \leq u \leq t$  in the finite function space. Using (4), its derivative is  $-e^{\lambda_r u} S_r(t-u)e_r$ . Integrating gives

$$u_r(t) := S_r(t)f_r - e^{\lambda_r t} f_r = \int_0^t e^{\lambda_r u} S_r(t-u)e_r du, \quad \|u_r(t)\|_\infty \leq \frac{256t}{K_r}. \quad (5)$$

The sign and absence of any factor 1/2 follow from the displayed original generator. The last inequality uses  $\lambda_r < 0$ , contraction, and (4). It is uniform in the full finite state and every  $t$  in a fixed compact interval. No common Hilbert realization, Gibbs projectivity, or limiting generator-domain theorem was used.

Independently,  $u'_r = L_r u_r + e^{\lambda_r t} e_r$  and  $u_r(0) = 0$ . For any fixed angle  $\theta$ , put  $v_\theta = \text{Re}(e^{-i\theta} u_r)$ . At a maximum over the finite state space,  $L_r v_\theta \leq 0$ ; the forcing is at most  $256/K_r$ . The finite comparison principle, or its upper-right Dini-derivative version at a change of maximizer, gives  $v_\theta(x, t) \leq 256t/K_r$ . Taking the supremum over all  $\theta$  gives (5), without a spurious  $\sqrt{2}$  loss. This is an alternative full-operator derivation, not a numerical enumeration or an exact-eigenfunction assumption.

### 4. One fixed positive time and arbitrarily late pairs

Pullback does not increase the supremum norm. Consequently, on every fine state,

$$\begin{aligned} D_{r,s}(t) &= (e^{\lambda_s t} - e^{\lambda_r t})f_s + u_s(t) - I_{r,s}u_r(t), \\ |D_{r,s}(t)(x)| &\geq |e^{\lambda_s t} - e^{\lambda_r t}| - 256t(K_s^{-1} + K_r^{-1}). \end{aligned} \quad (6)$$

If  $p_{r+2} \equiv 2$  or  $3 \pmod{5}$ , inverse multiplication swaps the residue classes  $\{1, 4\}$  and  $\{2, 3\}$ , hence swaps the two values in (4). At the preregistered original time  $t_* = 1/10$ , their exponential gap is

$$2e^{-1} \sinh(\sqrt{5}/5) \geq \frac{4}{15}, \quad (7)$$

because  $e < 3$ ,  $\sqrt{5} \geq 2$  and  $\sinh x \geq x$  for  $x \geq 0$ . When  $K_s \geq K_r \geq 768$ , the error in (6) is at most  $1/15$ . Thus every such flipping pair satisfies  $|D_{r,s}(t_*)(x)| \geq 1/5$  everywhere.

The required pairs occur beyond every threshold by the following elementary argument, reproduced from R-573. For  $B \geq 11$ , let  $P_B = \prod_{p \leq B, p \neq 5} p$ , and choose  $j \in \{1, 2, 3, 4\}$  with  $jP_B \equiv 1 \pmod{5}$ . Then  $1 + jP_B$  is two modulo five and one modulo every other prime at most  $B$ . All its prime factors exceed  $B$ , and at least one is two or three modulo five: products of residues one and four alone cannot give two. Given  $R$ , take  $B = p_{\max(R,4)+1}$  and one such prime  $p_k > B$ ;  $r = k - 2$ ,  $s = r + 1$  satisfy  $r \geq \max(R, 4)$  and the flip condition. Also  $K_r \geq K_4 = 2310 > 768$ . No finite prime table proves this quantifier.

Since the original fine Gibbs weights sum to one, (6)–(7) give

$$\forall R \exists s = r + 1 > r \geq R : \|D_{r,s}(1/10)\|_{L^2(\pi_s)}^2 \geq \frac{1}{25}. \quad (8)$$

Thus (1) is DISPROVED, already at  $T = 1/10$  (and every larger  $T$ ), with one fixed observable and  $\delta = 1/5$ . The obstruction is not confined to a Gibbs-negligible set. At  $t = 0$  the defect is zero; a derivative mismatch was not substituted for a finite-time lower bound.

## 5. Applicability and Devil’s-advocate review

The preregistered literature search used Norris, *Markov Chains*, chapter 2, section 2.1, Q-matrices and matrix exponentials, doi:10.1017/CBO9780511810633.004. This is finite background only; no inaccessible theorem or infinite-dimensional convergence result is imported. Contraction and variation of constants are derived above. R-570 applies to the exact finite generator/domain, R-571 to the unchanged pullback, and R-573 sections 2–4 to (3)–(4) and the prime argument. Their functional, regulator, state and time hypotheses are SATISFIED by the pinned tuple (2). Their executable independent/hostile/Lean evidence is replayed, but script PASS alone is not the analytic crosswalk. No Q3LOCK, TECT-YM, physical-time or external limit theorem is a premise. No novelty claim is made. The only new proposition is the passage from (4) to (8).

1. **Generator failure alone? DISMISSED.** Equation (5) supplies the uniform finite-time error; (7) is evaluated at a fixed positive time.
2. **Missing evolved roots? DISMISSED.** The Poisson expansion and independent maximum principle use the full  $L_r$ . Only initial increments vanish; the evolved character need not stay in a one-dimensional span.
3. **Sign, half factor, or time rescaling? DISMISSED.** Both eigenvalues are negative. The original generator has no half;  $1/10$  is a single original-time test value, not a cutoff-dependent clock.
4. **Fine/coarse error or measure mismatch? DISMISSED.** Both remainders occur in (6). Pullback uses only its supremum contraction; (8) uses the original normalized fine weights, not an isometric state transport.
5. **Finite table or moving observable? DISMISSED.** Equation (3) is one fixed-base observable; the Euclid argument supplies every tail. Non-flipping residues and zero time are negative controls, not witnesses.
6. **Hardcoded or approximate constants? DISMISSED.** The scripts recompute the edge count, rate remainder, exact fifth-root values, rational slack and threshold from pinned inputs. The constants  $\pi < 4$  and  $e < 3$  are analytic bounds; two directed Arb precisions are arithmetic controls.
7. **Full Lean or external referee? UPHELD as a boundary.** Five Lean theorems verify exponential/slack/triangle/weight/tail bridges only. The concrete finite Markov, vector integral, source and

prime arguments remain analytic. Independent code and hostile review have same-task authorship; external review is invited.

8. **Whole PAH or physical no-go? UPHOLD as a boundary.** This refutes the unchanged comparison's universal original-time full-cylinder convergence route. It does not refute R-570, R-571, every PAH model, state projectivity, or an independently authorized physical interpretation.

## 6. Result footer and reproducibility

Result ID: R-574; exploration EXP-001743.

Precise statement: Equation (8) refutes SG-001 at unchanged time 1/10, with original fine-Gibbs norm at least 1/5 beyond every cutoff threshold.

Scope: Exact R-573 PAH-v2 tuple, anchored square, full invariant counting algebra and original maps; local-state cutoff only.

Dependencies: Exact R-570 finite operators, R-571 comparison and R-573 character/remainder/prime proof; no imported infinite-limit theorem.

Evidence grade: ANALYTIC, EXACT, EXECUTED, INHERITED; partial Lean.

Reproduction command: Assemble the following displayed command on one line.

Expected output: Inherited R-573 and SG-001 primary, independent, hostile and partial Lean PASS; mathematical target DISPROVED.

Falsification gate: An exact source/rate/character, Markov contraction, vector integral, pullback or prime-tail error invalidates this negative proof.

Tier before / after: C6 stays T1; the T-054 active gate does not change.

No-overclaim statement: No new model, state, time, domain or comparison; no state-projectivity verdict, infinite-volume dynamics, continuum, physical Pre-A, Sector-A closure, spacetime, QFT, gravity, mass gap or TOE.

Next required action: Park this comparison route. Reenter only on an exact proof falsifier, or separately authorized evidence resolving a load-bearing assumption or supplying a physically motivated observable connection.

```
python -X utf8 verification/scripts/pah_v2_sg001_verify.py
--lean-cache E:/Dev/TECT/verification/lean/.lake/packages
--elan-home C:/Users/NaEun/.elan
```

No automatic weakened target or successor is queued. Brazovskii and horizon ideas remain separate, unadopted suggestions. One synthesis PDF is issued.